

Executive Summary: ESG and AI Deployments

By: Rania Siddiqui 07494

Introduction:

Environmental, Social, and Governance (ESG) principles are increasingly critical in evaluating the sustainability of AI deployments. In the context of AI, ESG focuses on understanding and mitigating the ecological footprint (energy use, emissions), ensuring ethical social impact, and upholding responsible governance practices. As AI models grow larger, responsible deployment practices become essential for a sustainable technological future.

Key findings:

- AWS is committed to achieving 100% renewable energy usage by 2025 [1].
- Amazon claims that AWS workloads are **up to 3.6 times more energy efficient** than typical enterprise data centers.
- Microsoft Azure pledges to be **carbon negative by 2030** and to remove all historical carbon emissions by 2050 [2].
- Azure reports its data centers already use **over 60% renewable energy** globally and aims for 100% renewable energy purchases by 2025.
- Google Cloud reduces environmental impact through renewable energy procurement and carbon reporting tools [3] and maintained a global average of 64% carbon-free energy across its data centers and offices in 2023 [4]
- Training **BERT Base** on a single TPU v3 pod reportedly consumes **~0.4 MWh** (400 kWh) energy [5].

Platform Comparison

While all three providers have sustainability goals, **Google Cloud** currently shows the most aggressive move toward carbon-free operations, particularly in select regions.

Azure leads in corporate carbon-negative commitments, while **AWS** has the largest operational scale but varies regionally in carbon intensity.

Recommendations

- **Prioritize Efficient Models:**
Use model optimization techniques like pruning, distillation, and quantization to reduce compute needs (e.g., using DistilBERT instead of BERT).
- **Choose Greener Regions:**
Deploy workloads in cloud regions with high renewable energy penetration (e.g., Oregon for AWS, Iowa for Google Cloud).
- **Time-Shift Workloads:**
Where possible, schedule non-urgent training during periods of higher renewable

energy availability (when such data is available).

- **Monitor and Offset Carbon Footprint:**

Leverage cloud provider dashboards (like Microsoft's Emissions Impact Dashboard) to track and compensate for emissions.

Conclusion

As AI models continue to scale, embedding ESG considerations into deployment strategies is crucial for reducing environmental harm. By combining model efficiency improvements, strategic region selection, and continuous monitoring, organizations can align AI innovation with sustainable development goals and ethical responsibility.

References:

1. Amazon Web Services, "AWS Sustainability," AWS Sustainability, 2025.
[Sustainable Cloud Computing - Amazon Web Services](#)
2. Microsoft Azure, Emissions Impact Dashboard
<https://www.microsoft.com/en-us/sustainability/emissions-impact-dashboard>
3. Google Cloud, "Cloud sustainability," Google Cloud, accessed Apr. 2025
[Sustainability | Google Cloud](#)
4. Google Sustainability, "Our goals for sustainable operations," Google Sustainability, accessed Apr. 2025
[Our Goals for Sustainable Operations - Google Sustainability](#)
5. <https://arxiv.org/abs/1906.02243>